

## PHP WARM-UP EXERCISE

### Prerequisite

- Basic HTML elements - buttons, headings, paragraph, form, links, list, table, and others.
- CSS - syntax, selector, background, border, margin, padding, units, position, and others.
- PHP - syntax, variable, operators, statements, loops, array, functions, post and get, pass by reference, methods, form handling, form validation, cookie, and session

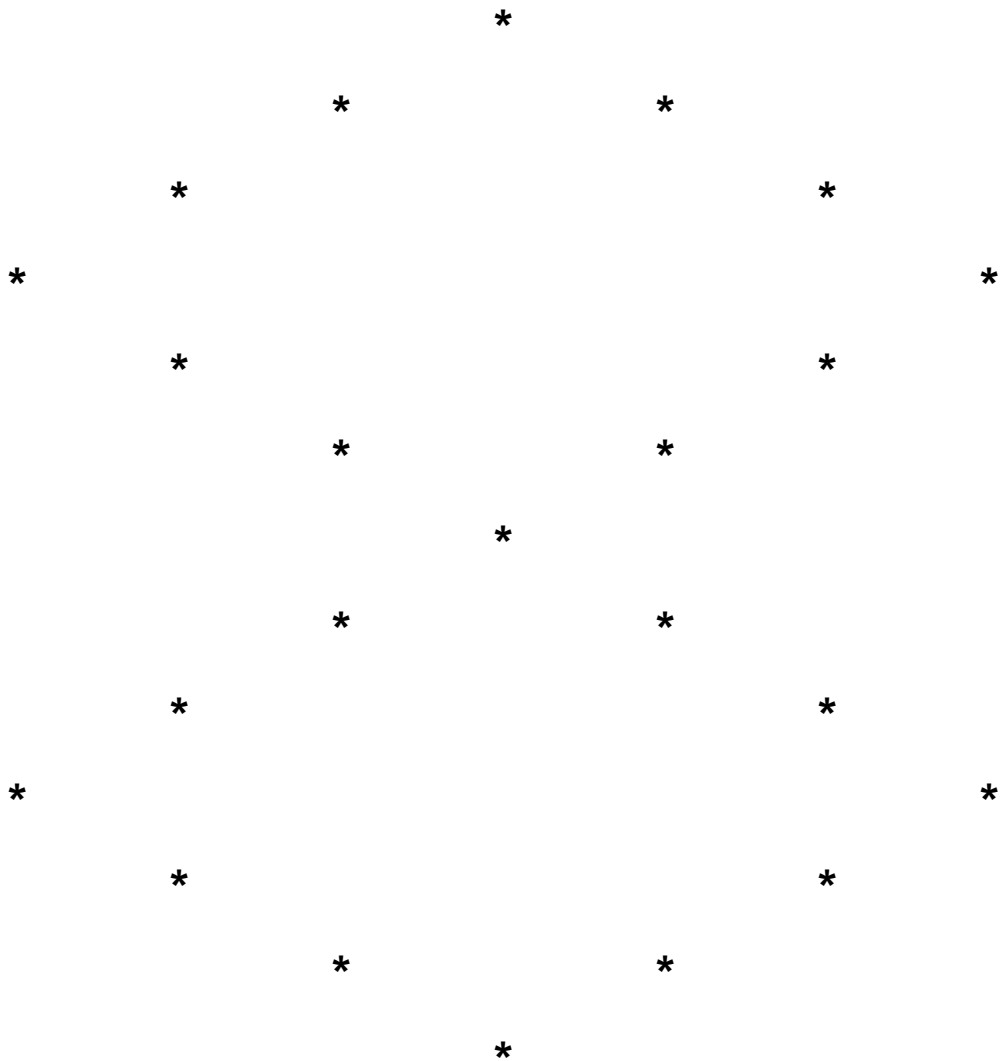
Note: Sharing of codes will automatically disqualify the applicant.

### Instruction

1. Create a folder
  - a. **Applicant name** as a folder name
  - b. Snake case format.
2. Programing language
  - a. Use vanilla PHP (no PHP framework like Laravel/Cakephp/Codeigniter is allowed)
3. Inside the folder, create a new **index.php** file
  - a. Use your name as header/title
  - b. Create a table of content of all the given tasks
  - c. All table items should redirect the user to task page
  - d. Use relative path to redirect (ex. href="/exercise1.php" )
4. Create a single stylesheet for all task pages
  - a. Make the page presentable as much as possible
  - b. Use of CDN CSS framework is allowed
5. Task page
  - a. Each task page should include the written task instruction/problem at the top
  - b. Use form and post method if the problem requires input from user
  - c. Add a home button that will redirect the user to index page
6. Make the code readable
  - a. Use sensible variable name
  - b. Proper indentation/spacing
  - c. Add a comment if necessary
7. Submit all the tasks in one folder at [patrick.v@agentsofvalue.com](mailto:patrick.v@agentsofvalue.com) and CC [hr\\_davaobbranch@agentsofvalue.com](mailto:hr_davaobbranch@agentsofvalue.com)

## Part 1: Fewer loops = better

1. Create the given pattern



2. Create the given pattern (x)

```
*
  5
    *
      3
        *
          1
        *
      3
    *
  5
*
  5
    *
      3
        *
          1
        *
      3
    *
  5
    *
      3
        *
          1
        *
      3
    *
  5
    *
```

3. Create the given pattern

```
1
2  4
3  6  9
4  8  12 16
5  10 15 20 25
4  8  12 16
3  6  9
2  4
1
```

4. Create the given pattern

|   |    |     |      |       |
|---|----|-----|------|-------|
| 1 | 2  | 4   | 8    | 16    |
| 2 | 6  | 18  | 54   | 162   |
| 3 | 12 | 48  | 192  | 768   |
| 4 | 20 | 100 | 500  | 2500  |
| 5 | 30 | 180 | 1080 | 6480  |
| 6 | 42 | 294 | 2058 | 14406 |

## Part 2

1. Generate a random character from a -k
  - Create a 4 x 5 table
  - Display all the random characters inside the table
  - Highlight all the character that belongs to the even column

(odd/even column)

|   |   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|---|----|
| a | b | c | d | e | f | g | h | i | j | k  |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |

Sample result

|   |   |   |   |   |
|---|---|---|---|---|
| A | C | E | G | h |
| f | A | j | K | A |
| f | h | l | j | d |
| b | b | b | b | b |

## 2. Manipulation of multidimensional array

- Create a 4x4 table
- Generate a random number from 1 - 100
- The number generated should always be unique
- Sum up the number per column and row

|           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|
| 1         | 2         | 3         | 4         | <b>10</b> |
| 5         | 6         | 7         | 8         | <b>26</b> |
| 9         | 10        | 11        | 12        | <b>42</b> |
| 13        | 14        | 15        | 16        | <b>58</b> |
| <b>28</b> | <b>32</b> | <b>36</b> | <b>40</b> |           |

## 3. Create a stack of integers using arrays( First in last out )

- create input fields and push a button to insert a new value
- Create a pop button to remove the top value
- Always display the existing stack content
- Do not use pre-defined PHP array functions like **array\_pop**

## 4. Create a queue of integers using arrays (first in first out )

- create input fields and push a button to insert a new value
- Create a pop button to remove the old value
- Always display the existing queue content
- Do not use pre-defined PHP array function like **array\_pop**

## 5. Using a PHP POST method, ask the user to input 2 numbers

- Create a Grid using 2 inputs as the length and width.
- Insert random CONSONANT letters into the grid
- Display the grid in table

Example

|   |   |   |   |
|---|---|---|---|
| b | c | d | f |
| g | s | r | t |
| h | m | n | p |
| k | l | q | q |

6. Fix the code function to output the correct result. **Explain your answer**

```
function bubbleSort($lists)
{
    $length = count($lists);

    for ($parent = 0; $parent < $length; $parent++) {

        for ($child = 0; $child < $length - $parent - 1; $child++) {

            if ($lists[$child] > $lists[$child + 1]) {
                $temp = $lists[$child + 1];
                $lists[$child] = $lists[$child + 1];
                $lists[$child + 1] = $temp;
            }
        }
    }
}
```